

CivicMysuru — Decision Log

HackMysuru 1.0, Phase 1 — Civic Governance & Clean Mysuru | Chosen sub-problem: Routing

1. Problem chosen and why

We chose the **Routing** sub-problem: automatically determining which authority a civic complaint belongs to, based on location and issue type. The brief made clear this is genuinely unresolved in Mysuru right now — the City Corporation is actively absorbing one City Municipal Council, four Town Panchayats, and eight Gram Panchayats, and even local leaders have publicly disagreed about which villages belong where. A system that assumes jurisdiction is fixed is already wrong.

2. Core design decision: versioned jurisdiction, not a static map

Every area-to-authority mapping is stored as a dated record (valid_from / valid_to), not a single fixed value. Routing always asks "which record was in force on the date this complaint was filed?" instead of reading a fixed dictionary. Adding or retiring a jurisdiction is a data change via an admin endpoint (POST /api/admin/jurisdiction-change) — never a code change or redeploy. We demonstrate this live: Hootagalli routes to its old Town Panchayat before 1 April 2026, and to Mysuru City Corporation from that date onward, using the exact same running server.

3. Handling the brief's other core realities

- **People will lie:** reused-evidence detection flags a photo filename reused across two different complaints and drops that complaint's verification score. Duplicate detection flags near-identical reports in the same area/issue category.
- **Not every complaint is equally urgent:** a keyword-driven priority engine scores each complaint (Critical/High/Medium/Low) based on hazard/urgency terms in the description.
- **Bad or tricky input:** an area with no matching jurisdiction record is routed to manual human review with an honest "UNASSIGNED" label, instead of a confident-sounding guess.
- **Bonus, not our chosen sub-problem — AI-assisted classification:** a standalone AI Detection page lets a user upload a photo and get an instant rule-based issue classification before filing a complaint. This is explicitly not the brief's "Detection Without Reporting" sub-problem, since it still requires a human to actively upload evidence rather than passively scanning footage — it's a genuinely useful bonus tool, not a claim to have solved a different sub-problem too.

4. Architecture

Frontend: static HTML/CSS/JS, no build step (pages: Home, Login, Report, Dashboard, AI Detection). Backend: FastAPI + SQLAlchemy + SQLite, with the routing/priority/duplicate logic in `routing_engine.py`, jurisdiction logic isolated in `jurisdiction_engine.py`, and the issue classifier isolated in `issue_classifier.py` so a real computer-vision model can later replace it without changing the API contract. The two halves communicate over a CORS-open REST API — no shared process, no hidden coupling.

5. Honest limitations (and the upgrade path for each)

- **Storage is SQLite, not a versioned GIS system.** Sufficient to prove the routing concept for this MVP; a real deployment would use PostGIS or similar for actual spatial boundaries.
- **Fake-evidence detection compares filenames, not image content.** A renamed reused photo would slip through today. The fix is a perceptual hash (pHash) of the actual image bytes once

real image upload (not just a filename string) is implemented — the swap only touches one function.

- **Duplicate detection uses word overlap, not GPS distance.** The report form already has a "detect my location" button that captures real latitude/longitude; wiring that into the duplicate check would meaningfully strengthen it.
- **The issue classifier is rule-based, not a trained vision model**, and says so honestly in its own output (note field) rather than being demoed as if it analyzes pixels.

6. How authority staff actually resolve a complaint

Every complaint carries a routed authority string (e.g. "Mysuru City Corporation"), but a routed label alone doesn't get a complaint in front of the right staff. The Dashboard's **authority filter** lets whoever is logged in narrow the shared complaint queue down to only the complaints routed to one specific office — simulating "log in as Nanjangud City Municipal Council staff and see only your queue." From there, the existing Start Work / Mark Resolved actions (PATCH `/api/complaints/{id}/status`) are how that office actually closes it out.

Honest scope of this: this is demo-authentication (any email/password), so the filter is a simulation of role separation, not enforced access control — anyone can still switch the dropdown to any authority. A real deployment would tie each staff account to exactly one authority server-side and never expose the others, rather than letting the client choose.

7. What we'd build next with more time

Real image upload + perceptual hashing for fake-evidence detection; GPS-distance-based duplicate detection; a proper GIS-backed jurisdiction table maintained by MCC staff through the existing admin endpoint; and a trained lightweight vision model behind the same classifier API contract, so the frontend needs no changes when it arrives.